

# Image of a periodic complex object in an optical system under partially coherent illumination

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Combined effects of amplitude and phase variations on the irradiance in the image of a periodic complex object, which has amplitude and phase distributions, have been investigated for an optical system with partially coherent object illumination. A general expression to obtain the illuminance in the image and the image contrast of such an object is derived by use of the concept of the effective source specifying the coherence condition. The irradiance in the image for the diffraction-limited aberration-free optical system illuminated with bounded effective sources that have uniform and nonuniform radiance, and also with annular illumination, have been calculated. Numerical calculations have been made to obtain the image, in which harmonics up to the thirteenth order are taken into account. One of the results is that abrupt amplitude and phase variations are extremely influential for the appearance of the image of a periodic, low-contrast, complex object for any mode of illumination. On the other hand, less abrupt changes of the amplitude and phase in a complex object produce less variation of the appearance and contrast of the image for any illumination mode except annular illumination. Images of pure amplitude and phase objects have also been obtained by use of the general treatment, and their characteristics are evaluated. It is suggested that a technique to manipulate the radiance distribution in the effective source is valuable in improving the fidelity of the image of a periodic complex object formed by an optical system under partially coherent illumination.

It is of importance, from the practical point of view, to study image formation in an optical system under partially coherent illumination, in which combined effects of the amplitude and phase distributions in a complex object on the image irradiance are considered. Knowledge of such combined effects could be required in cases that (i) the image of a biological specimen is ob-

served in a microscope, (ii) photographic optical density accompanied by a phase-changing relief image is measured with a microdensitometer, and (iii) partially coherent image processing by laser scanning<sup>1</sup> is carried out. In the imaging processes of the systems mentioned above, a specimen or an object with the amplitude and phase distributions is often illuminated with partially

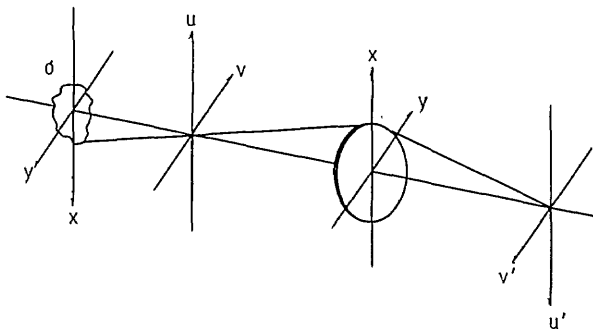


FIG. 1. Arrangement of the image forming system.

coherent light.

Although many papers have been published on image formation with partially coherent light, little work has been done about such combined effects. As far as we know, only the pure amplitude or pure phase object has been considered.<sup>2-9</sup> In a previous paper,<sup>10</sup> we derived the expression for the image irradiance of a sinusoidal complex object, and showed that the phase variation in an object was extremely influential for the appearance of the image and the image contrast. However, because of the nonlinear property of the partially coherent optical system, these results have provided information only about the combined effects of the amplitude and phase distributions in a sinusoidal complex object and were not applicable to the evaluation of image characteristics for other forms of complex objects.

In this paper, to overcome this defect, our treatment is generalized so that it can be applied to obtain the image irradiance of any periodic complex object. The general expression derived is used to investigate combined effects of the amplitude and phase distributions on the appearance of the image and its contrast for any complex object that has periodic structure. By representing the complex transmittance of the periodic complex object by a Fourier series, we can also calculate the image irradiance as a Fourier series.

For a diffraction-limited optical system, numerical calculation of the image can be made by replacing an infinite Fourier series by a finite series as long as we do not deal with an object containing extremely low-spatial-frequency components.

We will consider two types of object models which have both amplitude and phase distributions, i.e., a sinusoidal complex object and a trapezoidal complex object. Adjusting object parameters in these two models appropriately, we can express most complex objects that have periodic structures which involve abrupt and continuous amplitude and phase distributions. Fidelity of the image depends considerably upon combined effects of the magnitude of phase variation, the spatial phase distribution, contrast of the object, and the illumination mode. The images of objects that have the same contrast but different phase distributions, sometimes show quite different appearances to an observer. They appear to be the images of objects that have different contrast and fine structures. These phenomena occur frequently when we treat partially coherent imaging of

a low-contrast object that has large phase variations.

The image contrast, defined as the ratio of the Fourier coefficient in the illuminance in the image to the bias, is also calculated, and is compared with the appearance of the corresponding images.

A general expression for the image of a periodic complex object is obtained by modification of the formula originated in Ref. 1. To study the stated effects, the irradiance in the images is obtained under six different illumination conditions.

To avoid excessive mathematical derivation, we limit our study to the case of one dimension; however, the approach is easily applied to the two-dimensional case. In this paper, we mainly focus on the problem of partially coherent imaging of a biological specimen in a microscope.

## I. ANALYSIS

An image-forming system with four coordinate planes perpendicular to the optical axis is shown in Fig. 1. The coordinate systems are defined in the figure. The object  $O(u, v)$ , illuminated with the light wave radiated from the primary incoherent source  $\sigma$ , is imaged on the image plane. If system stationarity is assumed, the image irradiance is given by<sup>2</sup>

$$I(u') = \int \gamma(x) \left| \int o(s) f(x-s) e^{2\pi i s u'} ds \right|^2 dx, \quad (1)$$

where  $\gamma(x)$  is the effective source which characterizes the state of coherence of illuminating light,  $o(s)$  is the object spectrum, and  $f(s)$  represents coherent pupil function. Variables  $u, v; u', v';$  and  $x, y$  are expressed by reduced coordinates.<sup>10</sup> We may express the complex transmittance  $O(u)$  of a one-dimensional periodic complex object as a Fourier series

$$O(u) = \sum_n A_n e^{2\pi i n x_0 u}, \quad (2)$$

where  $x_0$  is the fundamental spatial frequency and  $A_n$  is the Fourier coefficient of component  $n x_0$ . Then, the object spectrum is given by

$$o(s) = \sum_n A_n \delta(s - n x_0), \quad (3)$$

where  $\delta$  denotes Dirac delta function. Substituting Eq. (3) into Eq. (1), and using the properties of Dirac delta function, the irradiance in the image becomes

$$I(u') = \sum_n \sum_m A_n A_m^* T(n, m) e^{2\pi i (n-m)x_0 u'}, \quad (4)$$

where the asterisk means complex conjugate. In Eq. (4),  $T(n, m)$  is the transmission cross coefficient<sup>11</sup> defined by

$$T(n, m) = \int \gamma(x) f(x + n x_0) f^*(x + m x_0) dx \equiv T_r(n, m) + i T_i(n, m), \quad (5)$$

where  $T_r(n, m)$  and  $T_i(n, m)$  are the real and imaginary parts of  $T(n, m)$ , respectively. According to the definition in Eq. (5), we obtain the relations

$$T_r(n, m) = T_r(m, n), \quad T_i(n, m) = -T_i(m, n), \quad (6)$$

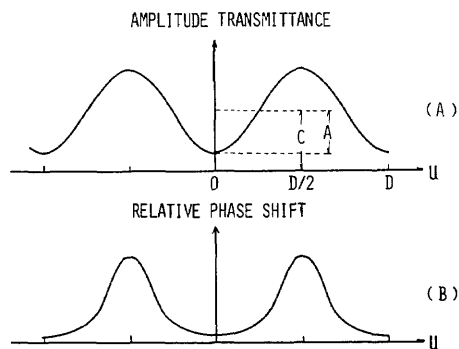


FIG. 2. Model of a sinusoidal complex object; (a) amplitude transmittance,  $C + A \cos 2\pi x_0 u$ , and (b) phase distribution,  $\exp(iB \cos 2\pi j x_0 u)$ .

in which  $\gamma(x)$  is assumed to be real. Set  $n - m = k$  for all values of  $n$  and  $m$  in Eq. (4), and put

$$A_n = R_n + iI_n, \quad A_m^* = R_m - iI_m, \quad (7)$$

where  $R_n$  and  $I_n$  are the real and imaginary parts of  $A_n$ . Applying the relations in Eqs. (5)–(7) to Eq. (4) with the rearranged series ( $n - m = k$ ), interchanging sums for  $n$  and  $k$ , and coupling exponentials, we may rewrite Eq. (4) as

$$I(u') = \sum_{k=0}^{\infty} (D_k \cos 2\pi k x_0 u' + D'_k \sin 2\pi k x_0 u') \\ = \sum_{k=0}^{\infty} E_k \cos(2\pi k x_0 u' - \theta_k), \quad (8)$$

where

$$E_k = \frac{1}{2}(D_k^2 + D_k'^2), \quad (8a)$$

$$\theta_k = \tan^{-1}(D'_k/D_k), \quad (8b)$$

$$D_k = \sum_n [P_{nk} T_r(n, n-k) + Q_{nk} T_i(n, n-k)], \quad (8c)$$

$$D'_k = \sum_n [Q_{nk} T_r(n, n-k) - P_{nk} T_i(n, n-k)], \quad (8d)$$

$$P_{nk} = \epsilon_k (R_n R_{n-k} + I_n I_{n-k}), \quad (8e)$$

and

$$Q_{nk} = \epsilon_k (R_n I_{n-k} - I_n R_{n-k}), \quad (8f)$$

where  $\epsilon_0 = 1$  and  $\epsilon_k = 2$  ( $k \neq 0$ ). In Eq. (8),  $D_k$  and  $D'_k$  are Fourier coefficients, and  $E_k$  and  $\theta_k$  indicate the amplitude and the image shift of  $k$ th harmonic in the irradiance in the image.

Equations (8) are the general expression for the image irradiance of a periodic complex object formed by an optical system under partially coherent illumination, in which the state of coherence and the characteristics of the image-forming system are condensed in the transmission cross coefficient, and the object characteristics are described by the parameters  $P_{nk}$  and  $Q_{nk}$ , respectively. Once the shape of the object is determined,  $R_n$  and  $I_n$  in Eqs. (8e) and (8f) for any  $n$  can be obtained easily. Although  $I(u')$  in Eqs. (8) is represented by double sums for  $n$  and  $k$ , it can be evaluated by calculating a finite number of terms for  $n$  and  $k$  in the case of a diffraction limited optical system.

## II. OBJECT MODELS

To investigate the stated effects, two kinds of object models that have both amplitude and phase distributions are considered.

One is the complex object that has sinusoidal amplitude transmittance  $C + A \cos 2\pi x_0 u$  and phase distribution  $\exp(i \sum_{j=1}^N B_j \cos 2\pi j x_0 u)$  shown in Fig. 2. Then, the complex transmittance of the object is given by

$$O(u) = (C + A \cos 2\pi x_0 u) \exp\left(i \sum_{j=1}^N B_j \cos 2\pi j x_0 u\right), \quad (9)$$

where  $C$ ,  $A$ , and  $B_j$  are constants. Parameters  $A$  and  $C$  specify the object contrast and the  $B_j$ 's characterize the phase variations. This object model is adequate to describe a sinusoidal density distribution recorded on photographic film accompanied by the phase-changing relief image caused by nonlinear modulation due to density variations. Using the Bessel-function formula

$$e^{iB_j \cos 2\pi j x_0 u} = \sum_n i^n J_n(B_j) e^{2\pi i n j x_0 u}, \quad (10)$$

the Fourier coefficient  $A_n$  of the complex transmittance of the object is calculated as

$$A_n = \sum_{j_1} \sum_{j_2} \cdots \sum_{j_N} i^{n-j_2-2j_3-\cdots-(N-1)j_N} J_{j_2}(B_2) J_{j_3}(B_3) \cdots J_{j_N}(B_N) \\ \times \{C J_{n-\alpha}(B_1) + \frac{1}{2} A i [J_{n+1-\alpha}(B_1) - J_{n-1-\alpha}(B_1)]\}, \quad (11)$$

$$\alpha = 2j_2 + 3j_3 + \cdots + Nj_N,$$

where  $J_n(B)$  is the Bessel function of the first kind and order  $n$ . If  $N=1$ , then

$$A_n = i^n C J_n(B_1) + \frac{1}{2} A i^{n+1} [J_{n+1}(B_1) - J_{n-1}(B_1)]; \quad (12)$$

and if  $N=2$ , then

$$A_n = \sum_m i^{n-m} J_m(B_2) \{C J_{n-2m}(B_1) + \frac{1}{2} A i [J_{n-2m+1}(B_1) - J_{n-2m-1}(B_1)]\}. \quad (13)$$

Details on the image contrast in the case of  $N=1$ , have been discussed in the previous paper.<sup>10</sup> However, for large  $N$ , numerical evaluation becomes impractical even though a large computer is used, because  $A_n$  is expressed by multiple sums, consisting of a number of cross terms of Bessel functions.

Another model is a periodic complex object with a period  $D$  that has trapezoidal amplitude transmittance and phase distributions shown in Fig. 3, in which the relative phase shift is assumed to be proportional to the amplitude distribution. The object characteristics are determined with seven parameters,  $A$ ,  $B$ ,  $C$ ,  $s_1$ ,  $s_2$ ,  $s_3$ , and  $s_4$  in the figure, where  $A$  and  $C$  are concerned with the contrast of the object,  $B$  with the phase variation, and  $s_1 \sim s_4$  with the form of the object. This model is suitable for assessment of the quality of the image of a complex object formed by a practical image-forming system such as a microscope, because most fine structures and characteristics in a practical object can be described by combinations of seven parameters.

The real and imaginary parts of a trapezoidal complex object with a fundamental frequency  $x_0$  ( $=1/D$ ) are calculated to be

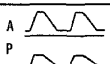
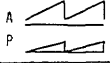
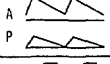
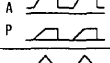
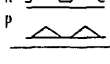
$$\begin{aligned}
R_n = & [A s_1 \operatorname{sinc}(2 n s_1)] + \left\{ (A s_2 - C s_1) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_2 + 2 n s_1)\right] \cos\left[\frac{1}{2} \pi(B - 2 n s_2 - 2 n s_1)\right] \right. \\
& + \frac{(C - A)[s_2 \sin \pi(B - 2 n s_2) + s_1 \sin 2 \pi n s_1]}{\pi(B - 2 n s_2 + 2 n s_1)} - \frac{(C - A)(s_2 - s_1) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_2 + 2 n s_1)\right] \sin\left[\frac{1}{2} \pi(B - 2 n s_2 - 2 n s_1)\right]}{\pi(B - 2 n s_2 + 2 n s_1)} \left. \right\} \\
& + [C(s_3 - s_2) \operatorname{sinc} n(s_2 - s_3) \cos \pi(n s_2 + n s_3 - B)] + \left\{ (C s_4 - A s_3) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_3 + 2 n s_4)\right] \cos\left[\frac{1}{2} \pi(B - 2 n s_4 - 2 n s_3)\right] \right. \\
& + \frac{(A - C)[s_4 \sin 2 \pi n s_4 + s_3 \sin \pi(B - 2 n s_3)]}{\pi(B - 2 n s_3 + 2 n s_4)} - \frac{(A - C)(s_3 - s_4) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_3 + 2 n s_4)\right] \sin\left[\frac{1}{2} \pi(B - 2 n s_4 - 2 n s_3)\right]}{\pi(B - 2 n s_3 + 2 n s_4)} \left. \right\} \\
& + [A(1 - s_4) \operatorname{sinc} n(1 - s_4) \cos \pi n(1 + s_4)], \quad (14a)
\end{aligned}$$

$$\begin{aligned}
I_n = & [-A s_1 \operatorname{sinc}(n s_1) \sin \pi n s_1] + \left\{ (A s_2 - C s_1) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_2 + 2 n s_1)\right] \sin\left[\frac{1}{2} \pi(B - 2 n s_2 - 2 n s_1)\right] \right. \\
& - \frac{(C - A)[s_2 \cos \pi(B - 2 n s_2) - s_1 \cos 2 \pi n s_1]}{\pi(B - 2 n s_2 + 2 n s_1)} + \frac{(C - A)(s_2 - s_1) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_2 + 2 n s_1)\right] \cos\left[\frac{1}{2} \pi(B - 2 n s_2 - 2 n s_1)\right]}{\pi(B - 2 n s_2 + 2 n s_1)} \left. \right\} \\
& + [C(s_2 - s_3) \operatorname{sinc} n(s_2 - s_3) \sin \pi(n s_2 + n s_3 - B)] + \left\{ (C s_4 - A s_3) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_3 + 2 n s_4)\right] \sin\left[\frac{1}{2} \pi(B - 2 n s_4 - 2 n s_3)\right] \right. \\
& - \frac{(A - C)[-s_4 \cos 2 \pi n s_4 + s_3 \cos \pi(B - 2 n s_3)]}{\pi(B - 2 n s_3 + 2 n s_4)} + \frac{(A - C)(s_3 - s_4) \operatorname{sinc}\left[\frac{1}{2}(B - 2 n s_3 + 2 n s_4)\right] \cos\left[\frac{1}{2} \pi(B - 2 n s_4 - 2 n s_3)\right]}{\pi(B - 2 n s_3 + 2 n s_4)} \left. \right\} \\
& + [-A(1 - s_4) \operatorname{sinc} n(1 - s_4) \sin \pi n(1 + s_4)], \quad (14b)
\end{aligned}$$

where  $\operatorname{sinc} x = \sin \pi x / \pi x$ . Adjusting seven object parameters appropriately, periodic complex objects with useful forms are determined, as shown in Table I. By use of Eq. (14), the real and imaginary parts,  $R_n$ 's and  $I_n$ 's, of spectra of these objects are easily obtained. They are listed in Table I.

If  $A = 0$  and  $B \neq 0$  in Eq. (11) or  $A = C$  and  $B \neq 0$  in Eq. (14), the model represents the pure phase object that has a periodic phase variation; and if  $A \neq C$  and  $B = 0$ , it becomes the pure amplitude object without phase change. Therefore, the formation of the image of a pure, periodic amplitude or phase object in a partially coherent optical system can be considered as a special case of the general treatment. Of course, analysis and numerical calculation become simpler than for a complex object.

TABLE I. Object models with useful forms and Fourier coefficients of them.  $R_n$  and  $I_n$  mean the real and imaginary parts of Fourier coefficients  $A_n$ .

| OBJECT MODEL                                                                       | OBJECT PARAMETERS                                         | $A_n$ (FOURIER COEFFICIENT)                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | $A, B, C, s_1 = P_1$<br>$s_2 = P_2, s_3 = P_3, s_4 = P_4$ | $R_n$ : EQ. (14a)<br>$I_n$ : EQ. (14b)                                                                                                                                                                                                                                       |
|  | $A, B, C$<br>$s_1 = s_2 = P_1, s_3 = s_4 = P_2$           | $R_n$ : $1 \operatorname{sinc}[\ ] + 3 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14a)<br>$I_n$ : $1 \operatorname{sinc}[\ ] + 3 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14b)                                                         |
|  | $A, B, C,$<br>$s_1 = 0, s_2 = s_3 = s_4 = 0$              | $R_n$ : $2 \operatorname{sinc}[\ ]$ IN EQ. (14a)<br>$I_n$ : $2 \operatorname{sinc}[\ ]$ IN EQ. (14b)                                                                                                                                                                         |
|  | $A, B, C,$<br>$s_1 = 0, s_2 = s_3 = P_1, s_4 = 0$         | $R_n$ : $2 \operatorname{sinc}[\ ] + 4 \operatorname{sinc}[\ ]$ IN EQ. (14a)<br>$I_n$ : $2 \operatorname{sinc}[\ ] + 4 \operatorname{sinc}[\ ]$ IN EQ. (14b)                                                                                                                 |
|  | $A, B, C,$<br>$s_1 = P_1, s_2 = P_2, s_3 = s_4 = 0$       | $R_n$ : $1 \operatorname{sinc}[\ ] + 2 \operatorname{sinc}[\ ] + 3 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14a)<br>$I_n$ : $1 \operatorname{sinc}[\ ] + 2 \operatorname{sinc}[\ ] + 3 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14b) |
|  | $A, B, C, s_1 = P_1$<br>$s_2 = s_3 = P_2, s_4 = P_3$      | $R_n$ : $1 \operatorname{sinc}[\ ] + 2 \operatorname{sinc}[\ ] + 4 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14a)<br>$I_n$ : $1 \operatorname{sinc}[\ ] + 2 \operatorname{sinc}[\ ] + 4 \operatorname{sinc}[\ ] + 5 \operatorname{sinc}[\ ]$ IN EQ. (14b) |

$P_1, P_2, P_3, \text{ AND } P_4 \leq 1$

### III. IMAGE CONTRAST

In general, the flux transmittance of a periodic complex object, that has both amplitude and phase distributions, and the irradiance in the corresponding image formed by an optical system under partially coherent illumination, contain infinite numbers of harmonics. Unfortunately, there is no linear relation between the spatial frequency components in the object and the resultant image, because partially coherent optical systems are nonlinear for either complex amplitude or irradiance. Moreover, the image irradiance involves, sometimes, frequency components which do not exist in the flux transmittance of the original object due to nonlinear modulation of an object. Therefore, we cannot define a useful measure such as the transfer function in the incoherent case to evaluate system performance in partially coherent illumination. Hence, to evaluate the characteristics of the image of a periodic complex object in a partially coherent optical system, the terms

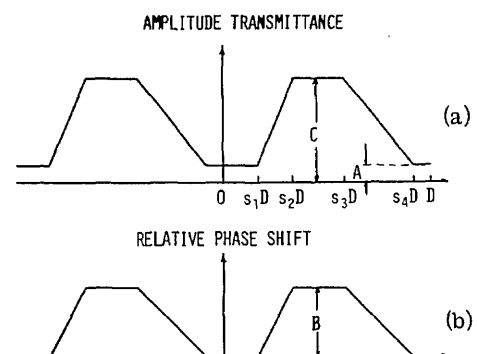


FIG. 3. Model of a trapezoidal grating-like complex object; (a) amplitude transmittance, and (b) phase distribution.  $A, B, C, s_1, s_2, s_3,$  and  $s_4$  are parameters describing the object characteristics.  $D$  is the period.

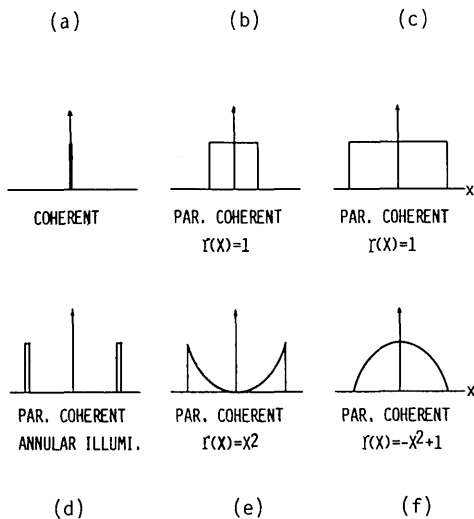


FIG. 4. Six effective sources.

image contrast and image shift have been applied, and are defined as the ratio of the Fourier coefficient of  $k$ th harmonic to the bias in the irradiance in the image

$$CT_k = E_k/E_0, \quad \theta_k = \tan^{-1}(D'_k/D_k). \quad (15)$$

Referring to Eqs. (8),  $CT_k$  and  $\theta_k$  may be specified by the object and system characteristics and are calcu-

- |                                                           |                                                      |
|-----------------------------------------------------------|------------------------------------------------------|
| (a) $\gamma(x) = \delta(x)$                               | (coherent source);                                   |
| (b) $\gamma(x) = 1$ for $ x  \leq 0.5 (R)$                | (bounded effective source with uniform radiance);    |
| $= 0$ otherwise,                                          |                                                      |
| (c) $\gamma(x) = 1$ for $ x  \leq 1.0 (R)$                | (bounded effective source with uniform radiance);    |
| $= 0$ otherwise,                                          |                                                      |
| (d) $\gamma(x) = 1$ for $0.95 (R') \leq  x  \leq 1.0 (R)$ | (annular source);                                    |
| $= 0$ otherwise,                                          |                                                      |
| (e) $\gamma(x) = x^2$ for $ x  \leq 1.0 (R)$              | (bounded effective source with nonuniform radiance); |
| $= 0$ otherwise,                                          |                                                      |
| (f) $\gamma(x) = -x^2 + 1$ for $ x  \leq 1.0 (R)$         | (bounded effective source with nonuniform radiance), |
| $= 0$ otherwise                                           |                                                      |

where  $R$  is the size of the effective source and is equivalent to the ratio of the condenser numerical aperture to the objective numerical aperture.  $R$  plays an important role in describing the coherent condition of the illuminating light.

$D_k$  and  $D'_k$  in Eq. (8) have to be calculated as infinite series of the product of the transmission cross coefficient  $T(n, n-k)$  and the combined terms of the real and imaginary parts of the object spectrum, in which the transmission cross coefficient  $T(n, n-k)$  for any  $n$  and  $k$ , can be evaluated by the overlapping area of three mutually shifted functions  $\gamma(x)$ ,  $f(x+nx_0)$ , and  $f^*[x+(n-k)x_0]$ . Fortunately, the numerical calculation of the above infinite series can be made by taking sums of finite terms for limited  $n$  unless the extremely low-spa-

lated for any reduced spatial frequency less than 2. Image contrast can be plotted in a way similar to the optical transfer function of an incoherent optical system. It should be noted that the image contrast and its shift serve to specify only the characteristics of the image of the stated complex object in a partially coherent optical system and not to describe the system performance. In spite of this restriction, they are still useful for evaluating the effects of nonlinearity of the system and nonlinear modulation of the object.

#### IV. NUMERICAL CALCULATION

Using relations in Eqs. (8) and (15), we calculate the image illuminance of a periodic complex object and image contrast in the optical system under partially coherent illumination. Let us consider a diffraction-limited aberration-free optical system and complex objects of periodic structures illuminated with bounded and annular effective sources. Hereafter, we only treat a symmetrical pupil function around the optical axis. That is, the pupil function of the image forming system is

$$f(x) = 1 \quad \text{for } |x| \leq 1 \\ = 0 \quad \text{otherwise.} \quad (16)$$

The effective sources are specified by six configurations depicted in Fig. 4, which are

tial-frequency component is treated, because the transmission cross coefficient has nonzero value for  $n, k$  (positive) and spatial frequency  $x_0$  satisfying  $0 \leq kx_0 \leq 2$ ,  $|(n-k)x_0 + 1| \leq R$ , and  $|nx_0 - 1| \leq R$ , otherwise it becomes zero.

#### V. APPEARANCE AND CONTRAST OF IMAGE

##### A. Square gratinglike complex object

A square gratinglike object that has amplitude and phase distributions is used to examine how the abrupt amplitude and phase variations in the object affected the image appearance and contrast. Object parameters are set to be  $s_1 = s_2 = 0.25$  and  $s_3 = s_4 = 0.75$ . The fundamental frequency of the object is selected as  $x_0 = 0.18$ , and

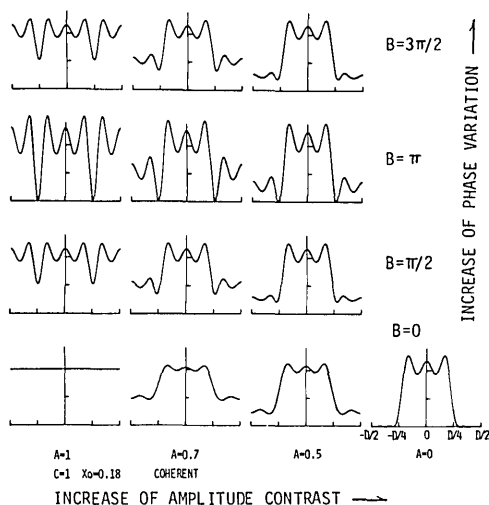


FIG. 5. Images of square gratinglike complex objects that have different amplitude and phase variations illuminated with coherent source in Fig. 4(a). Object parameter  $A$  specifying the amplitude contrast is changed, from the left to the right, as 1.0, 0.7, 0.5, and 0 and the parameter  $B$  specifying the phase variation is changed, from the bottom to the top, as 0,  $\frac{1}{2}\pi$ ,  $\pi$ , and  $\frac{3}{2}\pi$ , respectively.

a computer calculation has been made to get the image irradiance containing harmonics up to the eleventh order for integers in the range  $|n| \leq 25$ .

Figures 5–10 show the images under six coherence conditions specified by effective sources depicted in Fig. 4. In every figure, 13 images are drawn for their one period, and are arranged in accordance with changes of parameters  $A$  and  $B$  specifying the object amplitude contrast and the relative phase shift, respectively. They are, from left to right, 1 (pure phase object), 0.7, 0.5, and 0 for  $A$ , and from the bottom to the top, 0 (pure amplitude object),  $\frac{1}{2}\pi$ ,  $\pi$ , and  $\frac{3}{2}\pi$  for  $B$ , respectively. Further analysis shows that, for the diffraction-limited aberration-free optical system, images for  $B = \frac{3}{2}\pi$  coincide with those for  $B = \frac{1}{2}\pi$ . This arrange-

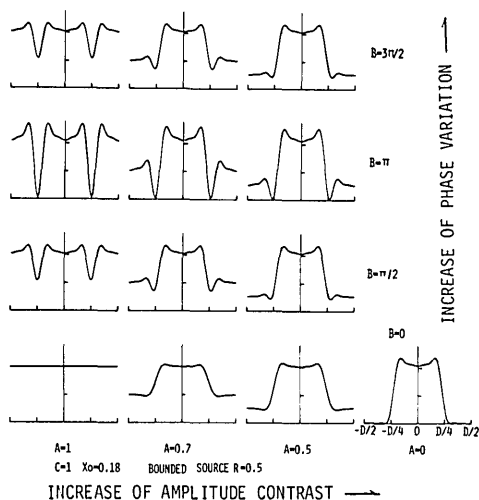


FIG. 6. Same as Fig. 5 but with the bounded effective source with uniform radiance with  $R=0.5$  in Fig. 4(b).

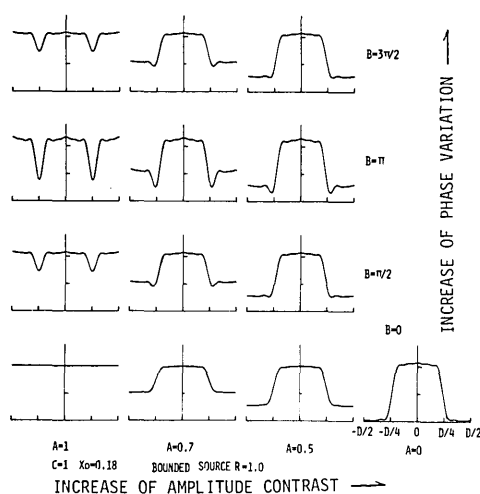


FIG. 7. Same as Fig. 5 but with the bounded effective source with uniform radiance with  $R=1.0$  in Fig. 4(c).

ment facilitates systematic evaluation of combined effects of amplitude and phase variations on the appearances of the images.

Figure 11 shows the image contrast of the images of the square gratinglike complex objects under six illumination conditions shown in Fig. 4, in which curves for the fundamental and second harmonic components are plotted for the objects whose parameters are  $A = 0.25$ ,  $C = 1$ , and  $B = 0, \frac{1}{2}\pi$ , and  $\pi$ , where the sign of the image contrast of the second harmonics is reversed. The image contrast of the second harmonic component provides information about nonlinear effects due to the partially coherent optical system and about nonlinear modulation of the object, because there is no harmonic components of the even order in the ideal image of the square gratinglike complex object.

Careful observation of Figs. 5–11 reveals the following significant facts and phenomena related to combined effects in the specified illumination modes:

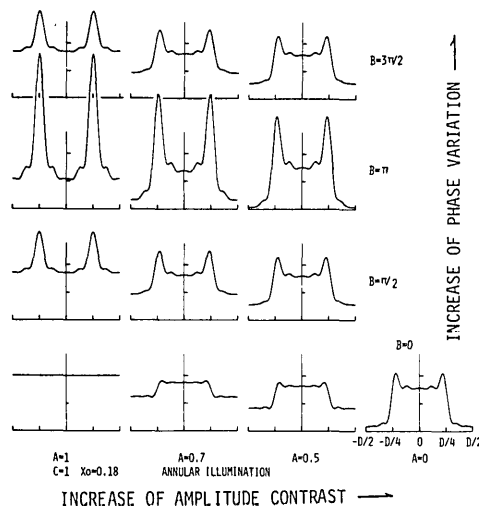


FIG. 8. Same as Fig. 5 but with the annular source in Fig. 4(d).

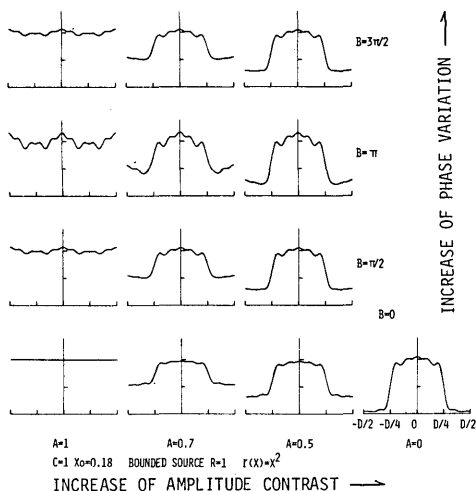


FIG. 9. Same as Fig. 5 but with the bounded effective source with nonuniform radiance distribution specified by  $\gamma(x) = x^2$  in Fig. 4(e).

(i) In the image formation of the diffraction-limited aberration-free optical system under coherent, near coherent, and annular illumination, phase change is extremely influential for the appearances of the images of the pure phase object and the low-amplitude contrast object that have sharp boundaries of phase. Most noticeable phenomena are generation of heavy ringing and sharp notches or peaks at the location of sharp boundaries.

(ii) There is a rapid decrease in the amount of ringing in the image and a monotonic decrease in edge gradient with decreasing coherence, as would be expected. On the contrary, a decrease in the depth of the notches is insensitive to decreasing coherence.

(iii) Increase of amplitude contrast and phase contrast in the complex object serve to increase the overall image contrast in any illumination condition. Increase of phase variation serves to sharpen the edge appearance in partially coherent illumination.

(iv) The structures of the images of the complex objects illuminated with the bounded effective source with nonuniform radiance distribution specified by  $\gamma(x) = x^2$  are much improved from the point of view of high-fidelity imaging of a complex object. There are no steep irradiance peaks or sharp notches at the location of sharp boundaries of amplitude and/or phase. On the other hand, the bounded source with nonuniform radiance distribution specified by  $\gamma(x) = -x^2 + 1$  does not serve to improve the image quality.

The phenomena mentioned above may be illustrated by examining the features in the images of the edge objects, that have sharp boundaries of amplitude and/or phase, illuminated with the annular source with various source sizes.

Let us consider the edge object referred to as the step function, positioned with the edge at the origin. The complex transmittance of this object is represented by the function

$$O_o(u) = \begin{cases} 1 & u > 0, \\ \frac{1}{2}(1 + Ae^{iB}) & u = 0, \\ Ae^{iB} & u < 0. \end{cases} \quad (17)$$

This object has a complex amplitude Fourier spectrum given by

$$o_o(x) = \frac{1}{2}[(1 + Ae^{iB})\delta(x) - (1 - Ae^{iB})i/\pi x], \quad (18)$$

where  $u$  and  $x$  are expressed by reduced coordinate, and  $A$  and  $B$  are constants related to the amplitude contrast and phase variation in the edge object, respectively. Now, consider that the edge object, illuminated with the light radiated from two point sources located at  $x = R$  and  $x = -R$  (annular source in one dimension), is imaged by the diffraction-limited aberration-free optical system whose pupil function is specified by Eq. (16). The radiance of the effective source is given by

$$\gamma(x) = \frac{1}{2}[\delta(x - R) + \delta(x + R)]. \quad (19)$$

Substitution of the relations of Eqs. (16), (18), and (19) into Eq. (1) yields the irradiance of the diffraction-limited image of the edge object. This may be given by

$$I_e(u') = (1/4\pi^2)(\pi^2(1 + A^2 + 2A \cos B) - 2\pi(A^2 - 1) \times [\text{Si}(X_2) - \text{Si}(X_1)] + (1 + A^2 - 2A \cos B) \times \{[\text{Si}(X_2) - \text{Si}(X_1)]^2 + [\text{Ci}(X_2) - \text{Ci}(X_1)]^2\}), \quad (20)$$

where  $u'$  is also expressed by the reduced coordinate and  $\text{Si}(X)$  and  $\text{Ci}(X)$  are the sine and cosine integral functions, and where  $X_1 = 2\pi u'(-1 + R)$  and  $X_2 = 2\pi u'(1 + R)$ . Equation (20) gives the irradiance in the diffraction-limited image of the edge object under annular illumination at all values of  $u'$  excepting  $u' = 0$ , since  $\text{Ci}(0) = -\infty$ . The image irradiance at  $u' = 0$  may be evaluated from the integral of the complex amplitude of the object spectrum within the diffraction-limited aperture.<sup>12</sup> By doing so, the image irradiance at  $u' = 0$  is obtained by

$$I_e(u' = 0) = \frac{1}{4\pi^2} \left[ \pi^2(1 + A^2 + 2A \cos B) + (1 + A^2 - 2A \cos B) \ln^2 \left( \frac{1 + R}{1 - R} \right) \right]. \quad (21)$$

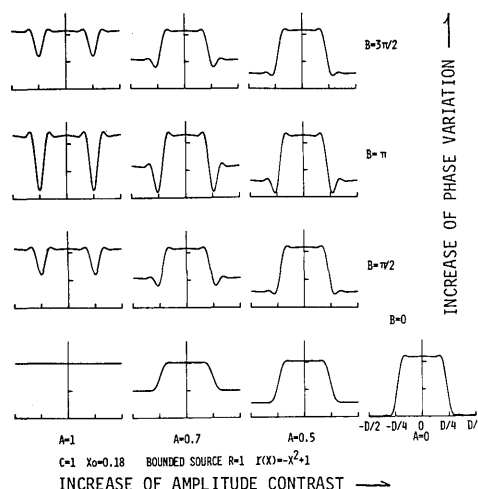


FIG. 10. Same as Fig. 5 but with the bounded effective source with nonuniform radiance distribution specified by  $\gamma(x) = -x^2 + 1$  in Fig. 4(f).

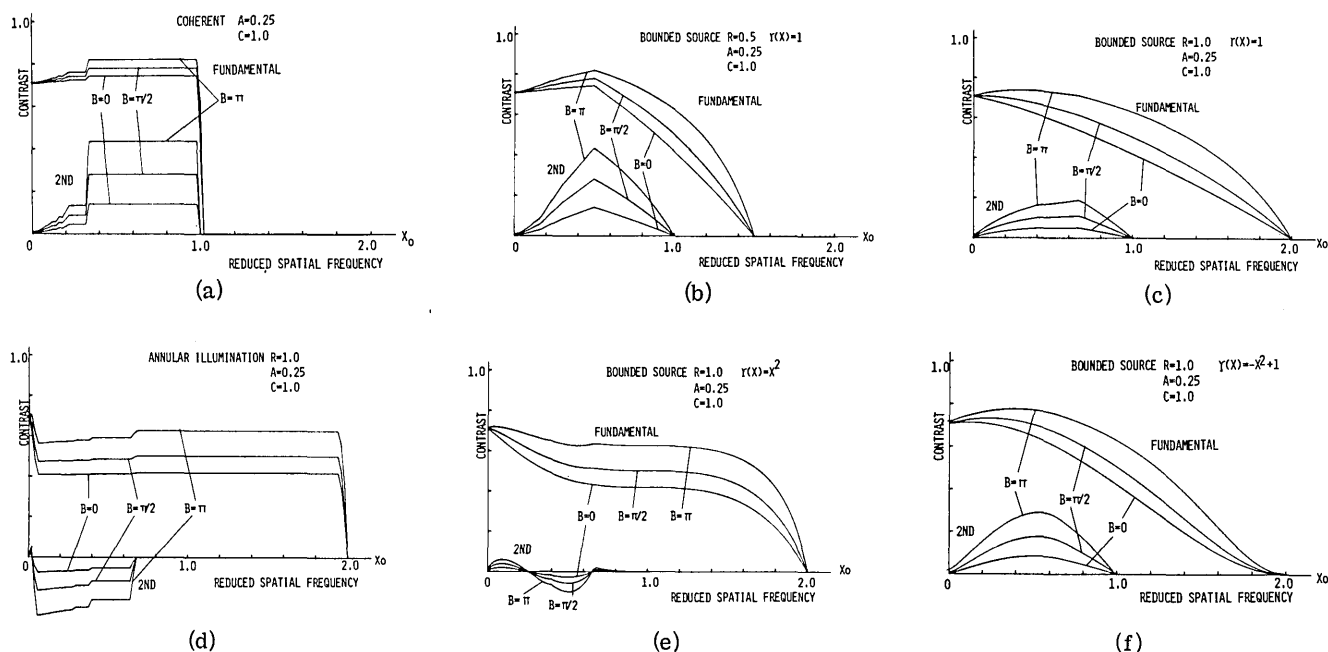


FIG. 11. Effects of phase variations on the image contrast for the fundamental and second harmonic components in the image of the square gratinglike complex object illuminated by six different effective sources in Fig. 4. Illuminating light sources to obtain contrast curves are: (a) the coherent source in Fig. 4(a); (b) the bounded effective source with uniform radiance with  $R=0.5$  in Fig. 4(b); (c) the bounded effective source with uniform radiance with  $R=1.0$  in Fig. 4(c); (d) the annular source in Fig. 4(d); (e) the bounded effective source with nonuniform radiance specified by  $\gamma(x)=x^2$ ; and (f) the bounded effective source with nonuniform radiance specified by  $\gamma(x)=-x^2+1$  in Fig. 4(f).

If  $R=0$ , Eqs. (20) and (21) give the image irradiance under central coherent illumination.

Generation of heavy ringing and sharp notches in coherent imaging can be interpreted by Gibbs phenomenon in the Fourier integral theory and illustrating diagram in Fig. 12. Figure 12 shows effects of phase changes on the appearance in the image of edge objects that have the same amplitude contrast, under coherent illumination [ $R=0$  in Eqs. (20) and (21)]. Patterns on the right- and left-hand sides in Fig. 12 indicate the behavior of the amplitude component of the diffraction-limited images of the edge objects and that of the corresponding image irradiance. The complex transmittance of the edge object is given by putting  $A=0.87$  in Eq. (17). Phase parameter  $B$  is changed for the figures from the top to the bottom as  $\pi$ ,  $\frac{1}{2}\pi$ , and  $0$ . It is evident from Fig. 12 that the notch goes down as the amplitude of the step increases and the maximum ringing and the deepest notches occur when  $B=\pi$ . That is, the amplitude of oscillation in ringing and the depth of the notch increase as the object amplitude contrast decreases and phase change is close upon  $\pi$ . The amount of oscillation in ringing is proportional to that of the step. A generation of the notch which goes to zero is due to the negative amplitude in the object. Thus, the diffraction-limited images of complex object that have sharp boundaries of the amplitude and/or phase contain heavy ringing and sharp dark lines (i.e., notches) at the location of sharp boundaries in the object. Quantitative dependence of these effects is clearly observed in the image structure diagram in Fig. 5.

Figure 13 shows the effect of variation of the source size in the annular source on a change in the appearance of the image of the pure phase object that has the sharp phase step of  $\pi$ . For this object, putting  $A=1$  and  $B=\pi$  in

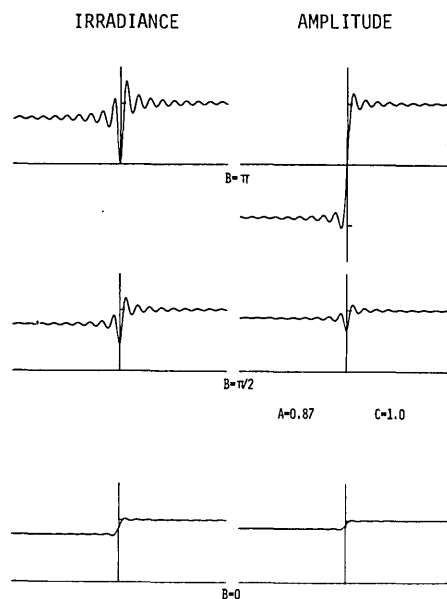


FIG. 12. Generation of the notches in the images of the edge objects that have sharp boundaries of amplitude and phase under coherent illumination. The parameter  $A$  specifying amplitude contrast in Eq. (17) is  $0.87$  for every figure, and the parameter  $B$  is  $\pi$ ,  $\frac{1}{2}\pi$ , and  $0$  for the top, middle, and bottom figures.

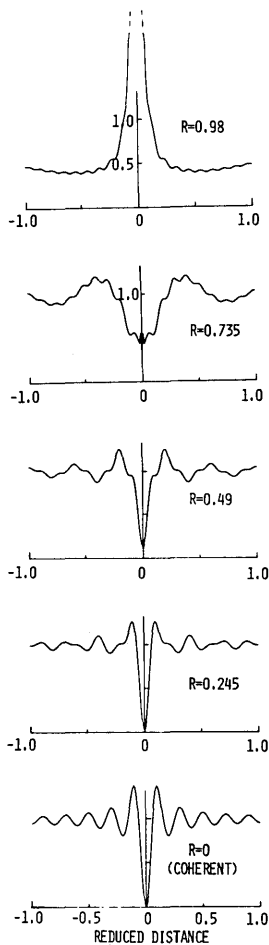


FIG. 13. Variation of the irradiance in the images of the pure phase object [ $A=1$  and  $B=\pi$  in Eq. (17)] that has sharp boundary of phase as a change in the source size  $R$  in the annular source.

Eqs. (20) and (21), the image irradiance is given by

$$I_o(u') = (1/\pi^2) \{ [\text{Si}(X_2) - \text{Si}(X_1)]^2 + [\text{Ci}(X_2) - \text{Ci}(X_1)]^2 \}, \quad (22)$$

$$I_o(u'=0) = (1/\pi^2) \ln^2 \left[ \frac{1+R}{1-R} \right]. \quad (23)$$

Figure 14 illustrates that variation of the source size in annular source leads to a change in the image irradiance at  $u'=0$ . From Fig. 14, image irradiance at  $u'=0$  increases abruptly as the source size is close upon unity, and it becomes infinity at  $R=1$ . Under these illumination condition, the sharp boundaries in the object may be encountered as the bright lines. The irradiance peaks at the location of sharp boundaries in the image structure diagram in Fig. 8 may be caused by this phenomenon.

Increase of the source size in the annular source corresponds to shift of the object spectrum in the pupil plane. If the source size exceeds unity, zero spatial frequency component is blocked out, and the resultant image appears in dark background. If the source size exceeds unity but is not so large, the sharp boundaries in the object can be observed as bright lines in dark background. This situation is similar to that in Schlieren method which obtains information about phase objects. Thus, the annular source, whose source size

is close upon unity, is worthy to obtain information about sharp boundaries of amplitude and/or phase in the complex object. However, it is not expected to get a high-fidelity image under such illumination condition, owing to the heavy shift of the object spectrum passing through the pupil.

The image of the edge object illuminated with the bounded effective source with uniform radiance can be evaluated from integrating the image irradiance in Eqs. (20) and (21) over the source area. A rapid decrease in the amount of ringing in the image with decreasing coherence may result from cancellation of the oscillating terms such as the sine and cosine integral functions during integration.

Now we explain the reason why the images of square gratinglike complex objects still contain sharp dips at the location of sharp boundaries in the object illuminated with the bounded effective source with uniform radiance ( $R=1$ ). To do so, we examine the image irradiance at the location of the sharp boundary ( $u'=0$ ) of the pure phase object, that has the phase step of  $\pi$ , under corresponding illumination condition. This can be given by integrating Eq. (22) over the source area ( $|R| \leq 1$ ), i. e.,

$$I_{R=1}(u'=0) = \frac{1}{\pi^2} \int_0^1 \ln^2 \left( \frac{1+R}{1-R} \right) dR. \quad (24)$$

The definite integral in Eq. (24) has the value of  $\frac{1}{3}\pi^2$ . Then,  $I_{R=1}(u'=0) = \frac{1}{3}$ . This value corresponds to the value of the dip in the image of the pure phase object ( $A=1$  and  $B=\pi$ ) on the right-hand side of the second row in Fig. 7.

From the fact that the annular source with  $R=1$  enables to enhance the edge structure, light radiated from the source rim of the bounded effective source mentioned above ought to enhance the image irradiance at the location of the sharp boundaries. On the other hand, that from the inner part of the source serves to produce the irradiance dip at the corresponding position ( $u'=0$ ) as shown in Fig. 14. In a quadrature, two effects may compensate each other. In the present case, the effect of the former is less than that of the latter. As a result, the sharp irradiance dips would occur at  $u'=0$ . This consideration and comparison of image structure diagrams in Figs. 7 and 8 lead to a conclusion to obtain

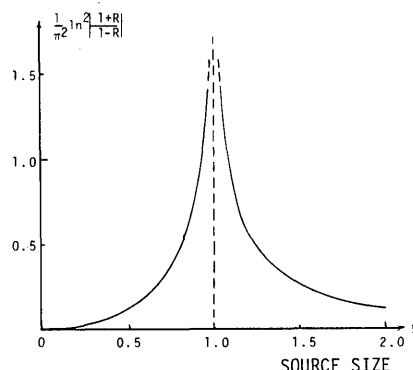


FIG. 14. Variation of the irradiance at the location of sharp boundary ( $u'=0$ ) in the image of the pure phase object used in Fig. 13 as a function of the source size in the annular source.

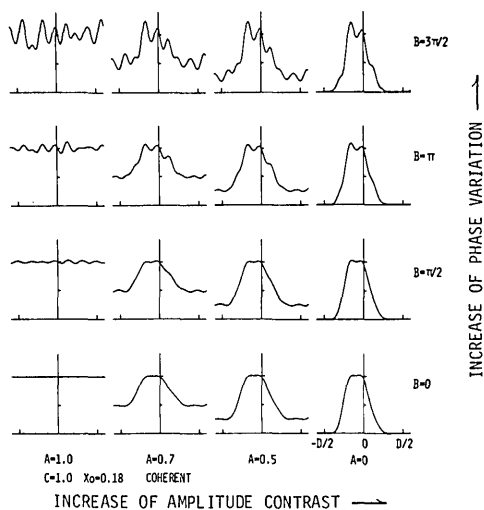


FIG. 15. Images of the trapezoidal gratinglike complex object illuminated with the coherent source in Fig. 4(a). Parameters  $A$  and  $B$  describe the amplitude and phase contrast in the object model in Fig. 3, where  $C=1$ .

the method to improve the quality of the image of the complex object, that has sharp boundaries of amplitude and/or phase. If the radiance in the inner part of the bounded source is reduced in a proper manner, maintaining high level around the source rim, the resultant image would provide the desirable image structures without sharp irradiance peaks or dips at the location of the sharp boundaries. Nonuniform radiance distribution specified by  $\gamma(x)=x^2$  depicted in Fig. 4(e) is such an example. Structures of the images in Fig. 9 obtained by using this source clarify that such nonuniform radiance distribution in the effective source gives desirable effects on suppression of generation of the irradiance peaks or dips.

As a matter of course, it is not expected to diminish the irradiance dips in partially coherent imaging when the complex object is illuminated with the effective source with nonuniform radiance distribution specified by  $\gamma(x)=-x^2+1$ . The images in Fig. 10 show that use of such effective source is meaningless to improve the image fidelity.

It is known that an optimum apodization to minimize the ringing effect can be attained by manipulation of the pupil function.<sup>13,14</sup> The facts mentioned above indicate that an optimum apodization to suppress the ringing effect and the irradiance peaks or dips at the location of the sharp boundaries in the object can be also made by changing radiance of the light source used in a nonlinear system. The optimum radiance distribution,  $\gamma_m(x)$ , of such an effective source is obtained by solving the following integral equation, i. e.,

$$\frac{1}{\pi^2} \int_0^1 \gamma_m(R) \ln^2 \left( \frac{1+R}{1-R} \right) dR = 1.$$

The dips appeared in the image structure diagrams in Figs. 5–7 looks like the first derivative of the sharp boundaries of amplitude and/or phase. However, it seems that they result from the effect of negative amplitude caused by the abrupt and fairly large phase change

at the edge rather than the effect of the first derivative of the sharp boundaries, because there is no sharp dips in the images of the pure amplitude objects that have sharp boundaries as shown in the bottom rows in Figs. 6 and 7.

## B. Trapezoidal gratinglike complex object

Figures 15 and 16 show the images of the complex objects that have trapezoidal amplitude and phase variations formed by an optical system illuminated with coherent source and the bounded effective source with uniform radiance distribution in Figs. 4(a) and 4(c), respectively. Form factors of the object in Fig. 3 are determined as  $s_1=0.1$ ,  $s_2=0.3$ ,  $s_3=0.5$ ,  $s_4=0.8$ , and  $C=1$ . The fundamental frequency is selected as  $x_0=0.18$ . The parameter  $A$  specifying the amplitude contrast of the object are changed, from the left to the right in the image structure diagram, as 1.0 (pure phase object), 0.7, 0.5, and 0 (pure amplitude object) and the parameter  $B$  specifying the phase variation of the object are changed, from the bottom to the top, as 0 (pure amplitude object),  $\frac{1}{2}\pi$ ,  $\pi$ , and  $\frac{3}{2}\pi$ , respectively. These numerical results reveal that the continuous phase change in the complex object has little influence on the image appearance and the image contrast depends only upon the object amplitude contrast in any illumination condition.

We also calculated the images of the sawtoothed objects that have amplitude and phase variations under six illumination conditions. There is no significant phenomenon present in the image structures that is not already essentially present in the image structures of the square gratinglike complex object and the trapezoidal gratinglike complex object.

## C. Sinusoidal complex object

Figure 17 shows dependence of the phase variation on the irradiance in the image of a sinusoidal complex object formed by an optical system illuminated with the

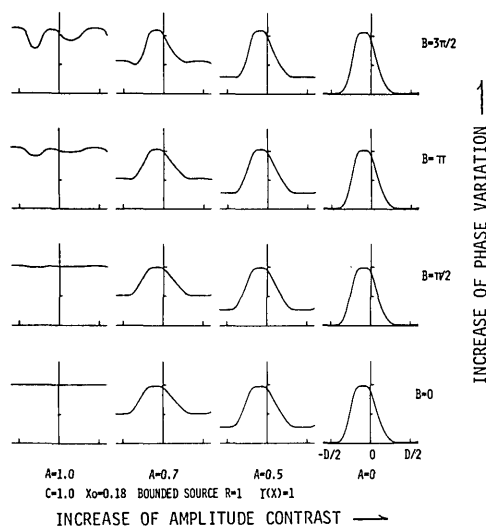


FIG. 16. Same as Fig. 13 but with the bounded effective source with uniform radiance with  $R=1$  in Fig. 4(c).

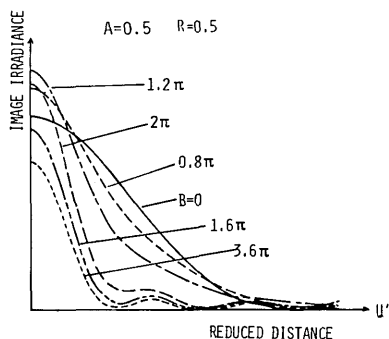


FIG. 17. Images of the sinusoidal complex objects formed by an optical system illuminated with the bounded effective source with uniform radiance with  $R=0.5$  in Fig. 4(b). The complex transmittance of the object is  $(1 + \frac{1}{2} \cos 2\pi x_0 u) \exp(iB \cos 2\pi x_0 u)$ .

bounded effective source with uniform radiance ( $R=0.5$ ). The complex amplitude of the original object is  $(1 + \frac{1}{2} \cos 2\pi x_0 u) \exp(iB \cos 2\pi x_0 u)$ . Individual curves are depicted for half-period of the resultant images of the complex objects with different phase distribution. From this figure, it becomes apparent that the image structure of the complex object that has sinusoidal amplitude and phase variations is distorted and the bright parts in the image are sharpened as the phase variation increases. This fact tells us that, in the course of microdensitometry of the photographic density accompanied by the phase changing relief image, phase variation due to the relief image must be compensated optically if we want to measure the precise density of the specimen.

## VI. SUMMARY AND CONCLUSION

By use of the Fourier-series analysis and the object models of the trapezoidal gratinglike complex object and the sinusoidal complex object, we have derived the general expression to obtain the image irradiance of any periodic complex object in an optical system under partially coherent illumination. Numerical results have clarified following significant phenomena in partially coherent imaging due to combined effects of amplitude and phase variations in the original object.

(i) The images of the complex objects that have sharp boundaries of amplitude and/or phase provide, sometimes, quite different appearances to an observer in accordance with difference of coherence conditions and phase variations in the objects, even though the amplitude contrast in the objects is the same. Therefore, when we observe the image of a low amplitude contrast complex object that has sharp boundaries, we must take into account that the resultant images tend to provide spurious appearance to an observer.

(ii) In partially coherent imaging of the complex object without sharp boundary of amplitude and/or phase, phase variation has little influence in changing the image structure and contrast.

(iii) Phase changes in the object become less important in changing the image structure as the amplitude contrast of the object increases. This conclusion is valid independent of the degree of coherence of the

illumination. This fact strongly suggests that when the image of the low-amplitude contrast complex object, such as a biological specimen, is observed in a microscope, it is desirable to dye the specimen if possible.

(iv) The structure of the image becomes obvious at low-amplitude and phase contrast levels in the object. An increase in object amplitude or phase contrast then serves only to increase the overall contrast of the image. There are no image structure features presented in the image of the highest contrast amplitude and phase objects which are not already essentially present in the image of the low-contrast amplitude and phase object.

(v) The bounded effective source whose radiance distribution is of the form  $\gamma(x) = x^2$  is worthy to suppress the ringing effect and notches in the images of the complex objects due to a nonlinear system. Effect of this source seems to be equivalent to that of the apodized aperture to minimize ringing. This fact suggests that there are better effective sources to provide high-fidelity image in partially coherent imaging of a complex object.

(vi) Annular illumination is useful to maintain high-image contrast to the incoherent resolution limit and to detect the sharp boundaries of amplitude and phase in the complex object. However, it is inadequate to form high-fidelity image of the complex object.

We mainly focused on the problem of evaluation of the image of a biological specimen observed in a microscope in this paper. The results are also applicable to other problems in microscopy. For instance, the results are useful during inspection of an integrated circuit in a microscope. In addition, we would like to point out that the results of this paper indicate that appropriate consideration must be paid to coherence effects on measurement of photographic optical density with a microdensitometer because variation of density in an emulsion is accompanied by a phase-changing relief.

Moreover, the results obtained also apply to partially coherent image processing by changing laser systems, because the image processing operations which can be implemented in a conventional partially coherent optical system such as that analyzed here can also be implemented by a scanning laser. The detector spatial sensitivity profile in the laser scanning system plays the same role as the radiance distribution of the incoherent source in the system treated in this paper.

We also point out that the results are applicable to the problem of image evaluation in electron microscopy to some extent, because in such a field observation of the low-amplitude contrast complex object with the high degree of coherence of illumination is a very important subject.

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